

# Week 1: Requirement Analysis and Customer Database Module

## 1. Requirement Analysis

### Understanding E-Commerce Business

An e-commerce system manages the complete process of buying, selling, and managing products online. The customer can search for products, add products to the cart, place orders, make payments, receive deliveries, and provide reviews or support requests.

### Process Flow

- **Browse / Search Product:** The customer searches for the required product.
- **Add to Cart:** The selected product is added to the cart.
- **Order Placed:** The customer selects the delivery address and places the order.
- **Checkout / Payment:** The customer proceeds to payment and completes the checkout process.
- **Delivery:** After successful payment, the product is dispatched and delivered.
- **Review & Support:** After delivery, the customer can provide a review or request support.

## 2. Customer Table Creation

A customer table was created to store customer details such as customer ID, first name, last name, email, phone number, and account creation date.

```
SQL> CREATE TABLE customer(  
2     customer_id VARCHAR2(20) PRIMARY KEY,  
3     first_name  VARCHAR2(20),  
4     last_name   VARCHAR2(20),  
5     email       VARCHAR2(30),  
6     phone       NUMBER(10),  
7     created_at  DATE  
8 );
```

Table created.

### 3. Describe Customer Table

The structure of the customer table was verified using the DESC command. The table contains the following fields:

- Customer ID
- First Name
- Last Name
- Email
- Phone
- Created At

```
SQL> desc customer
```

| Name        | Null?    | Type         |
|-------------|----------|--------------|
| CUSTOMER_ID | NOT NULL | VARCHAR2(20) |
| FIRST_NAME  |          | VARCHAR2(20) |
| LAST_NAME   |          | VARCHAR2(20) |
| EMAIL       |          | VARCHAR2(30) |
| PHONE       |          | NUMBER(10)   |
| CREATED_AT  |          | DATE         |

### 4. Insert Values

Five customer records were inserted into the customer table with the following details:

- **C-2001 – Lira James**
- **C-2002 – Kenny Robert**
- **C-2003 – Kairo William**
- **C-2004 – Suxi Martin**
- **C-2005 – Bliss Thomas**

```
SQL> INSERT INTO customer
  2 (customer_id, first_name, last_name, email, phone, created_at)
  3 VALUES
  4 (
  5     'C-2001',
  6     'Lira',
  7     'James',
  8     'lira2001@edss.com',
  9     9123456780,
 10     TO_DATE('2024-03-15', 'YYYY-MM-DD')
 11 );

1 row created.
```

## 5. Display Customer Records

The inserted customer records were displayed from the customer table. The records include Lira, Kenny, Kairo, Suxi, and Blissy along with their email addresses, phone numbers, and creation dates.

```
SQL> SELECT * FROM customer;
```

| CUSTOMER_ID | FIRST_NAME | LAST_NAME | EMAIL               | PHONE      | CREATED_A |
|-------------|------------|-----------|---------------------|------------|-----------|
| C-2001      | Lira       | James     | lira2001@edss.com   | 9123456780 | 15-MAR-24 |
| C-2002      | Kenny      | Robert    | kenny2002@edss.com  | 9234567891 | 21-JUN-24 |
| C-2003      | Kairo      | William   | kairo2003@edss.com  | 9345678902 | 10-SEP-24 |
| C-2004      | Suxi       | Martin    | suxi2004@edss.com   | 9456789013 | 18-JAN-25 |
| C-2005      | Blissy     | Thomas    | blissy2005@edss.com | 9567890124 | 27-MAY-25 |

## 6. Update

The phone number of customer **C-2005 (Blissy)** was updated successfully.

```
SQL> UPDATE customer
2  SET phone = 9786543210
3  WHERE customer_id = 'C-2005';

1 row updated.
```

## 7. Conclusion

The Customer Database Module was successfully implemented using Oracle SQL. The customer table was created, customer records were inserted and displayed, and an existing customer record was successfully updated.